Dairy Campus Research Data Platform

Data Descriptor, Standardised Metadata Dictionaries and Calculation Methodology

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Abstract

Dairy Campus (Wageningen University & Research, Leeuwarden) operates a multi-sensor experimental dairy herd for research on nutrition, feed efficiency, enteric methane mitigation, animal welfare and precision livestock farming. This data descriptor documents the research data platform: 19 datasets comprising 507 variables, with their data types, units, measurement origin and derivation. Derived quantities computed in the SQL layer (water correction of roughage intake, dry matter intake, yield-weighted milk composition, behaviour budgets and lactation timing) are given in closed form.

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1 Facility and sensor infrastructure

Dairy Campus houses an experimental herd of approximately 500 Holstein Friesian cows. The platform continuously ingests data from the following systems:

- **Hokofarm RIC troughs** – individual roughage intake bins on load cells with RFID identification.
- **Trioliet Triomatic** – automated feed kitchen and feeding robot; loading and trough deliveries.
- **Hokofarm/SAC concentrate stations** – computerised dispensing of up to four concentrates (A–D).
- **GEA milking systems and smart gates** – milk yield, milking duration, body weight and routing.
- **Nedap Velos smarttags** – behaviour classification into eating, ruminating, inactive and other active.
- **C-Lock GreenFeed** – enteric CH_4 and CO_2 fluxes during voluntary bait-fed visits.
- **Weather station** – ambient temperature, humidity, wind, precipitation and radiation.

Laboratory drying-oven dry matter analyses and official milk recording (CRV/MPR) complement the sensor streams.

2 Calculation methodology

This section formalises the derived variables that are computed inside the SQL tables and views of the platform. Only calculations documented in the data dictionary are included; the variable names given in parentheses refer to the columns in Section 4. Time stamps are denoted t, masses M (kg), and the index d refers to a calendar day.

2.1 Roughage intake per visit and per day

The Hokofarm RIC troughs are mounted on load cells. The intake and duration of visit v, and the daily fresh intake of roughage fraction $f \in \{1,2,3,4\}$ (`rv1.Prod`–`rv4.Prod`), are

$$I_v = M_v^{\text{start}} - M_v^{\text{end}}, \qquad \Delta t_v = t_v^{\text{end}} - t_v^{\text{start}}, \qquad I_{f,d} = \sum_{v \in \mathcal{V}_{f,d}} I_v, \tag{1}$$

where $\mathcal{V}_{f,d}$ is the set of visits of a cow to troughs holding fraction f on day d.

2.2 Correction for water added in the feed kitchen

Water is added to the ration in the Trioliet Triomatic feed kitchen. Intake excluding added water (`rvf.ProdExclWater`, `opnMinWater`; view `VL_Totaal_sub_ric_opn_prod`) is

$$I_{f,d}^{\text{excl}} \;=\; \frac{M_{\text{loaded}} - M_{\text{water}}}{M_{\text{loaded}}}\; I_{f,d}, \tag{2}$$

with M_{loaded} the total mass loaded into the mixer and M_{water} the mass of water added to that batch.

2.3 Dry matter intake

Weekly dry matter contents DM_f (%) are determined gravimetrically in the drying oven (`VL_totaal_droogstoof`). Total daily dry matter intake (`tot.opnDs`, kg DM/d) is

$$\text{DMI}_d \;=\; \sum_{f=1}^{4} I_{f,d}^{\text{excl}}\, \frac{\text{DM}_f}{100}. \tag{3}$$

2.4 Concentrate feeding

For the computerised concentrate stations, the residual (not consumed) concentrate (`SumVoeRst`) follows from the programmed allowance (`SumVoeGft`) and the consumed amount (`SumVoeOpnm`):

$$C_d^{\text{rest}} = C_d^{\text{granted}} - C_d^{\text{intake}}. \tag{4}$$

2.5 Yield-weighted milk composition

Milk recording (CRV/MPR) concentrations are aggregated per ISO calendar week w as yield-weighted means (`mpr_week.*`; view `VL_Dier_Datum_Crv`):

$$\bar{c}_w = \frac{\sum_{j \in \mathcal{J}_w} c_j \, Y_j}{\sum_{j \in \mathcal{J}_w} Y_j}, \qquad c \in \{\text{fat, protein, lactose, urea, SCC}\}, \tag{5}$$

where $\mathcal{J}_w$ is the set of test milkings in week w, c_j the concentration and Y_j (kg) the milk yield of milking j.

2.6 Daily behaviour budgets

Nedap Velos intervals k carry a behaviour state $s_k \in \{0, 1, 2, 3\}$ (0 other active, 1 inactive, 2 ruminating, 3 eating). The daily time budget per state (`overig`, `nietactief`, `herkouwen`, `vreten`) and the total (`total`) are

$$T_{s,d} = \sum_{k \in \mathcal{K}_d} \Delta t_k \, \mathbf{1}[s_k = s], \qquad T_d = \sum_{s=0}^{3} T_{s,d}, \tag{6}$$

with $\mathcal{K}_d$ the intervals recorded on day d and $\mathbf{1}[\cdot]$ the indicator function.

2.7 Lactation and reproduction timing

For a record on date t (all quantities in days):

$$\text{DIM} = t - t_{\text{calving}} \qquad (\texttt{LacDgn}, \texttt{IMDim}, \texttt{DaysInLactation}) \tag{7}$$

$$A = t - t_{\text{birth}} \qquad (\texttt{AgeDays}) \tag{8}$$

$$t_{\text{exp. calving}} = t_{\text{insemination}} + 280 \qquad (\texttt{VrwKlfDat}, \texttt{ExpecCalvingDate}) \tag{9}$$

$$D_{\text{to calving}} = t_{\text{exp. calving}} - t \qquad (\texttt{VrwDgn}) \tag{10}$$

3 Dataset overview

Table 1 lists all datasets; names link to the corresponding data dictionary.

Table 1: Datasets of the Dairy Campus research data platform.

#	Dataset	Content	Variables
1	`TVL_total`	Master daily integrated cow summary	97
2	`Hokofarm_SAC_concentrates`	Automatic concentrate feeder transactions	42
3	`Hokofarm_RIC`	Roughage intake control (RIC) visit transactions	18
4	`TVL_RIC_intake_dm`	Daily roughage intake and dry matter integration	12
5	`Triomatic_RIC_load`	Trioliet Triomatic feeding robot loading records	46
6	`Triomatic_RIC_drop`	Trioliet Triomatic feeding robot trough deliveries	46
7	`GEA_gates`	GEA separation and smart gate passages	5
8	`GEA_milkings`	GEA milking transactions	13
9	`TMR_Trioliet_cowconnect_mix`	CowConnect mixer wagon rations	12
10	`Velos_behaviour`	Nedap Velos behaviour intervals	4
11	`Velos_behaviour_day`	Nedap Velos daily behaviour budgets	7
12	`TMR_Trioliet_cowconnect_drop`	CowConnect feed distribution drops	11
13	`DC_DM_content`	Laboratory dry matter content (drying oven)	16
14	`Greenfeed_summarized_data`	GreenFeed enteric emission summaries	24
15	`Greenfeed_feed_aggregrated`	GreenFeed bait feed drops	13
16	`Weatherstation`	Dairy Campus meteorological station	18
17	`Missing_data`	Data completeness audit matrix	17

#	Dataset	Content	Variables
18	Environment_barn_Animal_day	Barn environment and animal daily profile	52
19	Environment_barn_Animal_experim	Experimental cohort animal daily records	54

4 Data dictionaries

All dictionaries share an identical layout. *Origin* indicates whether a variable is measured by a sensor or laboratory, or derived in the SQL layer; derivations refer to the equations in Section 2. A dash (—) denotes information that is not documented.

4.1 Master daily integrated cow summary

Dataset: `TVL_total` | **Source table:** `tvl_totaal` | **Variables:** 97

Table 2: Data dictionary of `TVL_total`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AnimalServerId` Animal server number	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
`AntMlkA` Number of milkings afternoon	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`AntMlkEtm` Number of milkings 24 hours	integer	–	Measured	Amount of milkings in the last 24 hours. 2 is the assumed value. 1 or 3+ means something is off during morning and/or afternoon milking.	—
`AntMlkO` Number of milkings morning	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`bcs` Body condition score	integer	–	Measured	Automatic body condition score camera	—
`BegDatPrf` Starting date experiment	date	yy-mm-dd	Measured	Expected start date of the experiment	—
`BehCod` Treatment code	string	text	—	Abbreviaton of the treatname name. Set by the researcher.	—
`BehNam` Treatment name	string	text	—	Full name of the treatment. Set by the researcher	—
`Blk` Blocking group	integer	–	—	Number of the assigned block. Set by the researcher.	—
`C21-10` Total concentrate via station 1 till 4	number	kg	Derived	Dag totaal krachtvoer caroussel	—
`C21-9` Concentrate feed milking caroussel	number	kg	—	Kilogram of concentrate feed dispatched at the milking caroussel. NOT the uptake, that cannot be registered currently.	—
`CelStd` Somatic cell count standard	string	x1000 cells/mL	Derived	Somatic cell count of the monthly milk test	Official somatic cell count (SCC) determined from routine 4-weekly milk recording samples (MPR), analyzed by laboratory. *Source:* `T_Animal_Milk Samples (MlkCon. SCCDay)`
`Dat` Date	date	yy-mm-dd	Measured	Date of the data record	—
`DelPro-13` Feed from the milking robot	number	kg	—	Intake feed 1 from the milkingrobot	—
`DrgZetDat` Start date dry period	date	yy-mm-dd	Measured	Starting date of the dry period	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`EiwStd` Protein concentration standard	number	%	Derived	Protein concentration of the monthly milk test	Milk crude protein percentage determined from official routine milk recording laboratory analysis. *Source:* `T_Animal_Milk Samples (MlkCon. ProteinPercentage)`
`EndDatPrf` Ending date experiment	date	yy-mm-dd	Measured	Expected end date of the experiment	—
`GewA` Weight afternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
`GewO` Weight morning	number	kg	Measured	Weight of the animal after milking in the morning	—
`gf_visits` GreenFeed measurements	string	—	—	Measurements on cow emission (summarized data)	—
`gf_cupdrops` GreenFeed cupdrops	string	—	—	GreenFeed portions concentrate feed.	—
`Hok` Pen / Stable number	integer	–	—	Location of where the cows are housed.	—
`HokMin1` Pen / Stable number day before	integer	–	—	Location of where the cows are housed day before	—
`AtributedKVA` Attributed Concentrate A	number	kg	Derived	Total allocated/programmed amount of concentrate feed type A for the cow on the specified date.	Sum of programmed concentrate A quantities across feeding stations (Hokofarm / SAC). *Source:* `T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv`
`TotalIntakeKVA` Concentrate feed station 1	number	kg	Derived	Total intake feed from SAC box 1	Total consumed quantity of concentrate feed A by the animal at the automatic feeding station on the specified date. *Source:* `T_Raw_Hokofarm CD (totalIntakeKVA)`
`AtributedKVB` Attributed Concentrate B	number	kg	Derived	Total allocated/programmed amount of concentrate feed type B for the cow on the specified date.	Sum of programmed concentrate B quantities across feeding stations (Hokofarm / SAC). *Source:* `T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv`
`TotalIntakeKVB` Concentrate feed station 2	number	kg	Derived	Total intake feed from SAC box 2	Total consumed quantity of concentrate feed B by the animal at the automatic feeding station on the specified date. *Source:* `T_Raw_Hokofarm CD (totalIntakeKVB)`
`AtributedKVC` Attributed Concentrate C	number	kg	Derived	Total allocated/programmed amount of concentrate feed type C for the cow on the specified date.	Sum of programmed concentrate C quantities across feeding stations (Hokofarm / SAC). *Source:* `T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv`
`TotalIntakeKVC` Concentrate feed station 3	number	kg	Derived	Total intake feed from SAC box 3	Total consumed quantity of concentrate feed C by the animal at the automatic feeding station on the specified date. *Source:* `T_Raw_Hokofarm CD (totalIntakeKVC)`

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`AtributedKVD` Attributed Concentrate D	number	kg	Derived	Total allocated/programmed amount of concentrate feed type D for the cow on the specified date.	Sum of programmed concentrate D quantities across feeding stations (Hokofarm / SAC). *Source:* `T_Raw_Hokofarm CD / TVL_Totaal_sub_kv`
`TotalIntakeKVD` Concentrate feed station 4	number	kg	Derived	Total intake feed from SAC box 4	Total consumed quantity of concentrate feed D by the animal at the automatic feeding station on the specified date. *Source:* `T_Raw_Hokofarm CD (totalIntakeKVD)`
`HrhNum` Repeating number	integer	–	—	Number of repeat of a previously performed test	—
`InsDat` Insemination day	date	yy-mm-dd	Measured	Date of last insemination	—
`Jar` Year	integer	–	Measured	Year of the data record	—
`KetStd` Ketose concentration standard	integer	–	Derived	Ketose concentration of the monthly milk test	Ketosis diagnostic classification code (0: normal/negative, 1: suspected subclinical, 2: clinical ketosis). *Source:* `T_Animal_Milk Samples (MlkCon. KetosisCode)`
`KgMlkA` Kilogram milk afternoon	number	kg	Measured	Kilogram of milk registred at the afternoon milking event	—
`KgMlkEtm` Kilogram milk 24 hours	number	kg	Measured	Kilogram of milk in the last 24 hours	—
`KgMlkO` Kilogram milk morning	number	kg	Measured	Kilogram of milk registred at the morning milking event	—
`KlfDat` Calving date	date	yy-mm-dd	Measured	Date of last calving	—
`LacDgn` Days in lactation	integer	–	Derived	Amount of days producing milk	Date - calving date
`LacNum` Parity	integer	–	Measured	Parity (amount of calvings) of the animal	—
`LacStd` Lactose concentration standard	number	%	Derived	Lactose of the monthly milk test	Milk lactose percentage determined from official routine milk recording laboratory analysis. *Source:* `T_Animal_Milk Samples (MlkCon. LactosePercentage)`
`LevNum` Registration number	integer	text + number	—	National registration number of the animal.	—
`LftDgn` Age	integer	–	Measured	Age of the animal in days	—
`MlkStd` Milk standard	number	kg	Derived	Kilogram of milk produced during the standard (once a month) milk composition sampling	Total daily milk yield recorded during the official milk control recording day. *Source:* `T_Animal_Milk Samples (MlkCon.Milk QuantityMass)`
`mpr_dag.Fat_ Concentration` Weekly milk fat concentration	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`mpr_dag.Lactose_Concentration` Weekly milk lactose concentration	number	%	Measured	Weekly milk lactose concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`mpr_dag.Protein_Concentration` Weekly milk protein concentration	number	%	Measured	Weekly milk protein concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`mpr_dag.SCC` Weekly somatic cell count	string	10^3 cells mL^-	Measured	Weekly somatic cell count during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`mpr_dag.Urea_Concentration` Weekly milk urea concentration	string	$mg\,dL^{-1}$	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`mpr_week.Fat_Concentration` Calculated day fat concentration	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	ISO-calendar week yield-weighted average milk fat concentration calculated across regular CRV/MPR test recordings: sum(Fat_ Concentration * MilkingYield) / sum(MilkingYield). Used to derive Fat- and Protein-Corrected Milk (FPCM). *Source:* `VL_Dier_Datum_Crv`
`mpr_week.Lactose_Concentration` Calculated day lactose concentration	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk lactose concentration calculated across regular CRV/MPR test recordings: sum(Lactose_ Concentration * MilkingYield) / sum(MilkingYield). *Source:* `VL_Dier_Datum_Crv`
`mpr_week.Protein_Concentration` Calculated day protein concentration	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk protein concentration calculated across regular CRV/MPR test recordings: sum(Protein_ Concentration * MilkingYield) / sum(MilkingYield). Used to derive Fat- and Protein-Corrected Milk (FPCM). *Source:* `VL_Dier_Datum_Crv`

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`mpr_week.SCC` Calculated day somatic cell count	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average somatic cell count (1,000 cells/mL) calculated across regular CRV/MPR test recordings: sum(SCC * MilkingYield) / sum(MilkingYield). *Source:* `VL_Dier_Datum_Crv`
`mpr_week.Urea_Concentration` Calculated day urea concentration	string	mg dL^{-1}	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk urea concentration (mg/dL) calculated across regular CRV/MPR test recordings: sum(Urea_ Concentration * MilkingYield) / sum(MilkingYield). *Source:* `VL_Dier_Datum_Crv`
`OorNum` Ear number	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
`OpnRvDsHans` Intake roughage dry matter Hans	string	—	—	Intake of dry matter from the roughage, Hans (Will be removed from table)	—
`PrdSta` Production stage	string	text	—	Identifying the stage of the cow, producing milk or not	—
`PrfCod` Experiment number	integer	–	—	Code assigned by Dairy Campus to identify the experiment. Can be the same as the MyProjects number, or differ.	—
`RowID` Row ID	integer	–	—	Unique number for row in a specific table.	—
`rv1.ds` Roughage 1 dry matter	string	—	—	Roughage 1 dry matter	—
`rv1.name` Roughage name 1	string	text	—	Name of the roughage treatment consumed by the animal	—
`rv1.opnDs` Roughage 1 intake dry matter	string	—	—	Roughage 1 intake dry matter	—
`rv1.Prod` Roughage product 1	number	kg	Derived	Total uptake in kilograms of the roughage	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
`rv1.ProdExclWater` Roughage 1 intake excluding water	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 1 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* `VL_Totaal_sub_ric_opn_prod`
`rv1.ricBez` Roughage 1 RIC bin visits	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
`rv1.trBatches` Roughage 1 batches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
`rv2.ds` Roughage 2 dry matter	number	kg	—	Roughage 2 dry matter	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`rv2.name` Roughage name 2	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
`rv2.opnDs` Roughage 2 intake dry matter	number	kg	—	Roughage 2 intake dry matter	—
`rv2.Prod` Roughage product 2	number	kg	Derived	Total uptake in kilograms of the roughage.	—
`rv2.ProdExclWater` Roughage 2 intake excluding water	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 2 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* `VL_Totaal_sub_ric_opn_prod`
`rv2.ricBez` Roughage 2 RIC bin visits	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
`rv2.trBatches` Roughage 2 batches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
`rv3.ds` Roughage 3 dry matter	number	kg	—	Roughage 3 dry matter	—
`rv3.name` Roughage name 3	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
`rv3.opnDs` Roughage 3 intake dry matter	number	kg	—	Roughage 3 intake dry matter	—
`rv3.Prod` Roughage product 3	number	kg	Derived	Total uptake in kilograms of the roughage.	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
`rv3.ProdExclWater` Roughage 3 intake excluding water	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 3 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* `VL_Totaal_sub_ric_opn_prod`
`rv3.ricBez` Roughage 3 RIC bin visits	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
`rv3.trBatches` Roughage 3 batches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
`rv4.ds` Roughage 4 dry matter	number	kg	—	Roughage 4 dry matter	—
`rv4.name` Roughage name 4	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
`rv4.opnDs` Roughage 4 intake dry matter	number	kg	—	Roughage 4 intake dry matter	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`rv4.Prod` Roughage product 4	number	kg	Derived	Total uptake in kilograms of the roughage.	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
`rv4.ProdExclWater` Roughage 4 intake excluding water	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 4 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* `VL_Totaal_sub_ric_opn_prod`
`rv4.ricBez` Roughage 4 RIC bin visits	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
`rv4.trBatches` Roughage 4 batches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
`tot.opnDs` Total intake drymatter	number	kg	Derived	Total uptake in kilograms of dry matter	Total daily dry matter intake (kg DM/day). Calculated by summing the dry matter intake of all roughage fractions: (OpnMinWater * DryMatterPercentage). *Source:* `VL_Totaal_sub_ric_opn_prod`
`tot.Prod` Total product	number	kg	—	Total roughage 1 t/m 4	—
`UreStd` Urea concentration standard	number	mg/100 gram	Derived	Urea concentration of the monthly milk test	Milk urea nitrogen content determined from official routine milk recording laboratory analysis. *Source:* `T_Animal_Milk Samples (MlkCon.Urea)`
`VetStd` Fat concentration standard	number	%	Derived	Fat concentration of the monthly milk test	Milk fat percentage determined from official routine milk recording laboratory analysis. *Source:* `T_Animal_Milk Samples (MlkCon.Fat Percentage)`
`VlgKlfDat` Next calving date	date	yy-mm-dd	Derived	Next calving date, not predicted, but added after calving	—
`VrwDgn` Expected number of days	integer	–	Derived	Number of days from date to expected calving date based on insemination date	Expected calving date minus current date. *Source:* `TVL_Dier_ Datum_Stal`
`VrwKlfDat` Expected calving date	date	yy-mm-dd	Derived	Expected calving date based on insemination date	Insemination date + 280 days
`Wek` Week	integer	–	Measured	Calender week number of data record	—
`WekDag` Day of the week	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—

4.2 Automatic concentrate feeder transactions

Dataset: `Hokofarm_SAC_concentrates` | **Source table:** `t_raw_hokofarmvb` | **Variables:** 42

Table 3: Data dictionary of `Hokofarm_SAC_concentrates`.

Variable	Type	Unit	Origin	Description	Derivation / source
`ashA` Ash A	number	kg	Derived	Ash, indigestible part of feed A	—
`ashB` Ash B	number	kg	Derived	Ash, indigestible part of feed B	—
`ashC` Ash C	number	kg	Derived	Ash, indigestible part of feed C	—
`ashD` Ash D	number	kg	Derived	Ash, indigestible part of feed D	—
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`crudeFiberA` CrudeFibre A	number	kg	Derived	Crude fiber in feed A	—
`crudeFiberB` CrudeFibre B	number	kg	Derived	Crude fiber in feed B	—
`crudeFiberC` CrudeFibre C	number	kg	Derived	Crude fiber in feed C	—
`crudeFiberD` CrudeFibre D	number	kg	Derived	Crude fiber in feed D	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`dateTime` Date and time	datetime	yyyy-mm-dd-hh-mm-ss	Measured	Date and time	—
`dryMatterA` Dry matter A	number	kg	Derived	Dry matter of feed A	—
`dryMatterB` Dry matter B	number	kg	Derived	Dry matter of feed B	—
`dryMatterC` Dry matter C	number	kg	Derived	Dry matter of feed C	—
`dryMatterD` Dry matter D	number	kg	Derived	Dry matter of feed D	—
`endTime` End time	number	hh-mm-ss	Measured	End time of action/behaviour	—
`energyA` Energy A	string	VEM	Derived	Energy content A	—
`energyB` Energy B	string	VEM	Derived	Energy content B	—
`energyC` Energy C	string	VEM	Derived	Energy content C	—
`energyD` Energy D	string	VEM	Derived	Energy content D	—
`fatA` Fat A	number	kg	Derived	Fat content A	—
`fatB` Fat B	number	kg	Derived	Fat content B	—
`fatC` Fat C	number	kg	Derived	Fat content C	—
`fatD` Fat D	number	kg	Derived	Fat content D	—
`feedTypeA` Feed type A	string	text	—	Type of feed A	—
`feedTypeB` Feed type B	string	text	—	Type of feed B	—
`feedTypeC` Feed type C	string	text	—	Type of feed C	—
`feedTypeD` Feed type D	string	text	—	Type of feed D	—
`fileName` File name	string	text	—	Name of a file	—
`hashValue` Hash value	integer	–	—	Validation record for data processing	—
`intakeA` Intake A	number	kg	Derived	Feed intake A	Calculated by endWeight - startWeight
`intakeB` Intake B	number	kg	Derived	Feed intake B	Calculated by endWeight - startWeight
`intakeC` Intake C	number	kg	Derived	Feed intake C	Calculated by endWeight - startWeight

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`intakeD` Intake D	number	kg	Derived	Feed intake D	Calculated by endWeight - startWeight
`proteinA` Protein A	integer	–	Derived	Protein content A	—
`proteinB` Protein B	integer	–	Derived	Protein content B	—
`proteinC` Protein C	integer	–	Derived	Protein content C	—
`proteinD` Protein D	integer	–	Derived	Protein content D	—
`startTime` Start time	number	hh-mm-ss	Measured	Start time of action/behaviour	—
`station` Station	integer	–	Measured	Station or location of activity	—
`totalTime` Total time	integer	s	Derived	Total duration	Calculated by endTime - startTime
`transponder` Transponder	integer	–	Measured	Transponder number of the cow	—

4.3 Roughage intake control (RIC) visit transactions

Dataset: `Hokofarm_RIC` | **Source table:** `t_raw_hokofarmvr` | **Variables:** 18

Table 4: Data dictionary of `Hokofarm_RIC`.

Variable	Type	Unit	Origin	Description	Derivation / source
`ash` Ash	number	kg	Derived	Ash, indigestible part of the feed	—
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`crudeFibre` CrudeFibre	number	kg	Derived	Crude fiber in the feed	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`datetime` Date and time	datetime	yyyy-mm-dd-hh-mm-ss	Measured	Date and time	—
`dryMatter` Dry matter	number	kg	Derived	Dry matter	—
`duration` Duration	integer	s	Derived	Duration of action/behaviour	Calculated by endTime - startTime
`endTime` End time	number	hh-mm-ss	Measured	End time of action/behaviour	—
`endWeight` End Weight	number	kg	Measured	End weight	—
`energy` Energy	string	VEM	Derived	Energy content	—
`fat` Fat	number	kg	Derived	Fat content	—
`feedType` Feed type	string	text	—	Type of feed	—
`intake` Intake	number	kg	Derived	Feed intake	Calculated by endWeight - startWeight
`protein` Protein	number	kg	Derived	Protein content	—
`startTime` Start time	number	hh-mm-ss	Measured	Start time of action/behaviour	—
`startWeight` Start weight	number	kg	Measured	Start weight	—
`station` Station	integer	–	Measured	Station or location of activity	—
`transponder` Transponder	integer	–	Measured	Transponder number of the cow	—

4.4 Daily roughage intake and dry matter integration

Dataset: `TVL_RIC_intake_dm` | **Source table:** `TVL_totaal_hokofarm_ds` | **Variables:** 12

Table 5: Data dictionary of `TVL_RIC_intake_dm`.

Variable	Type	Unit	Origin	Description	Derivation / source
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`dsPct` dry matter percentage	integer	–	Derived	Percentage of dry matter in the feed	—
`feedtype` feedtype	string	text	—	Type of feed	—
`name` Name	string	text	—	Name collected based on other value	—
`opnDsVR` Intake dry matter Roughage RIC	number	kg	—	Intake of dry matter from roughage RIC	—
`opnIncWater` Intake including water	number	kg	Derived	Feed intake including water	—
`opnMinWater` Intake subtracting water	number	kg	Derived	Feed intake excluding water	—
`opnProd` Intake product	number	kg	—	Total intake of product	—
`ricBez` Ric visits	integer	–	—	Number of visits to RIC	—
`RowID` Row ID	integer	–	—	Unique number for row in a specific table.	—
`trBatches` Tioliet batches	integer	–	—	Number of times the triolith has released feed.	—

4.5 Trioliet Triomatic feeding robot loading records

Dataset: `Triomatic_RIC_load` | **Source table:** `t_raw_trioliet` | **Variables:** 46

Table 6: Data dictionary of `Triomatic_RIC_load`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AnimalCount` Animal count	integer	–	—	The number of animals	—
`Date` Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
`Extra` Extra	integer	–	—	Extra information	—
`FeedingGroup` Feeding group	integer	–	Measured	Log source 0 = 1, log source 1 = NULL betekenis Feeding group?	—
`filename` Filename	string	text	—	Name of the file containing information	—
`FinishedReady` Finished ready	string	—	—	NULL?	—
`GroupID` Group ID	integer	–	—	Id of the group	—
`GroupName` Group name	string	text	—	Name of the group	—
`Id` ID	integer	–	—	ID	—
`IngredientName` Ingredient name	string	text	—	Name of the ingredient	—
`k42` k42	integer	–	—	Validation record for data processing	—
`LogSource` Log source	integer	–	—	0 meaning feed preparation, and 1 the distribution of feed to RIC bins	—
`NettoLoadedWeight` Netto loaded weight	number	kg	Measured	Netto weight loaded	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`NrOfSplittings` Number of splittings	integer	–	—	Number of splittings	—
`NumberRecipeLines` Number of recipe lines	integer	–	—	Total number of recipe lines	—
`OpdrachtID` Job ID	integer	–	—	Job ID	—
`PerformedFeedFactor` Performed feed factor	integer	–	—	Performed feed factor	—
`PerformedProductDryWeightKgAfterDrop` Kg dry weight after drop	number	kg	Measured	Kg dry weight after drop	—
`PerformedProductDryWeightPartAfterDrop` Part of dry weight after drop	number	kg	Measured	Part of dry weight after drop	—
`PerformedProductError` Performed product error	number	kg	Derived	Error, difference between Scheduled kg and performed kg feed	—
`PerformedProductKgAfterDrop` Kg product after drop	number	kg	Measured	Kg product after drop	—
`PerformedProductKgAtStop` Kg product at stop	number	kg	Measured	Kg product at stop	—
`PerformedTime` Performed time	number	hh-mm-ss	Measured	Performed time	—
`PushingGroup` Pushing group	integer	–	—	Pushing group	—
`Ready` Ready	integer	–	—	Ready	—
`ReadyTime` Ready time	number	hh-mm-ss	—	Ready time, if log source 1	—
`ReceptLineNr` Recept line number	integer	–	—	Current number of recipe line	—
`ReceptName` Recipe name	string	text	—	Name of the feed recipe	—
`ScheduledDryWeightContent` Scheduled dry weight content	integer	–	Derived	Scheduled dry weight content	—
`ScheduledDryWeightContentPart` Scheduled dry weight content part	integer	–	Derived	Scheduled dry weight content part	—
`ScheduledDryWeightKg` Scheduled dry weight kg	number	kg	Derived	Scheduled dry weight kg	—
`ScheduledFeedFactor` Scheduled feed factor	integer	–	—	Scheduled feed factor	—
`ScheduledPart` Scheduled part	integer	–	Derived	Scheduled part	—
`ScheduledProductKg` Scheduled product kg	number	kg	—	Scheduled product kg	—
`ScheduledTime` Scheduled time	number	hh-mm-ss	—	Scheduled time, if log source 1	—
`SplittedNr` Splitted number	integer	–	—	Splitted number, only 1	—
`StartPosition` Start position	integer	–	Measured	Start position, if subgroup not 0	—
`StartTime` Start time	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
`StartWeight` Start weight	number	kg	Measured	Start weight, if subgroup not 0	—
`StopPosition` Stop position	integer	–	Measured	Stop position, if subgroup not 0	—
`StopTime` Stop time	number	hh-mm-ss	Measured	Stop time, if subgroup not 0	—
`StopWeight` Stop weight	number	kg	Measured	Stop weight, if subgroup not 0	—
`StrawGroup` Straw group	integer	–	—	Straw group	—
`SubGroupNr` Subgroup number	integer	–	—	Number of the subgroup, 0 no subgroup	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`TakeoffTime` Take off time	number	hh-mm-ss	Measured	Take off time, if log source 1	—
`TrackSection` Track section	integer	–	—	Track section, if subgroup 1 if there is no subgroup 0	—

4.6 Trioliet Triomatic feeding robot trough deliveries

Dataset: `Triomatic_RIC_drop` | **Source table:** `t_raw_trioliet` | **Variables:** 46

Table 7: Data dictionary of `Triomatic_RIC_drop`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AnimalCount` Animal count	integer	–	—	The number of animals	—
`Date` Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
`Extra` Extra	integer	–	—	Extra information	—
`FeedingGroup` Feeding group	integer	–	Measured	Log source 0 = 1, log source 1 = NULL betekenis Feeding group?	—
`filename` Filename	string	text	—	Name of the file containing information	—
`FinishedReady` Finished ready	string	—	—	NULL?	—
`GroupID` Group ID	integer	–	—	Id of the group	—
`GroupName` Group name	string	text	—	Name of the group	—
`Id` ID	integer	–	—	ID	—
`IngredientName` Ingredient name	string	text	—	Name of the ingredient	—
`k42` k42	integer	–	—	Validation record for data processing	—
`LogSource` Log source	integer	–	—	0 meaning feed preparation, and 1 the distribution of feed to RIC bins	—
`NettoLoadedWeight` Netto loaded weight	number	kg	Measured	Netto weight loaded	—
`NrOfSplittings` Number of splittings	integer	–	—	Number of splittings	—
`NumberRecipeLines` Number of recipe lines	integer	–	—	Total number of recipe lines	—
`OpdrachtID` Job ID	integer	–	—	Job ID	—
`PerformedFeedFactor` Performed feed factor	integer	–	—	Performed feed factor	—
`PerformedProductDryWeightKgAfterDrop` Kg dry weight after drop	number	kg	Measured	Kg dry weight after drop	—
`PerformedProductDryWeightPartAfterDrop` Part of dry weight after drop	number	kg	Measured	Part of dry weight after drop	—
`PerformedProductError` Performed product error	number	kg	Derived	Error, difference between Scheduled kg and performed kg feed	—
`PerformedProductKgAfterDrop` Kg product after drop	number	kg	Measured	Kg product after drop	—
`PerformedProductKgAtStop` Kg product at stop	number	kg	Measured	Kg product at stop	—
`PerformedTime` Performed time	number	hh-mm-ss	Measured	Performed time	—
`PushingGroup` Pushing group	integer	–	—	Pushing group	—
`Ready` Ready	integer	–	—	Ready	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`ReadyTime` Ready time	number	hh-mm-ss	—	Ready time, if log source 1	—
`ReceptLineNr` Recept line number	integer	–	—	Current number of recipe line	—
`ReceptName` Recipe name	string	text	—	Name of the feed recipe	—
`ScheduledDryWeightContent` Scheduled dry weight content	integer	–	Derived	Scheduled dry weight content	—
`ScheduledDryWeightContentPart` Scheduled dry weight content part	integer	–	Derived	Scheduled dry weight content part	—
`ScheduledDryWeightKg` Scheduled dry weight kg	number	kg	Derived	Scheduled dry weight kg	—
`ScheduledFeedFactor` Scheduled feed factor	integer	–	—	Scheduled feed factor	—
`ScheduledPart` Scheduled part	integer	–	Derived	Scheduled part	—
`ScheduledProductKg` Scheduled product kg	number	kg	—	Scheduled product kg	—
`ScheduledTime` Scheduled time	number	hh-mm-ss	—	Scheduled time, if log source 1	—
`SplittedNr` Splitted number	integer	–	—	Splitted number, only 1	—
`StartPosition` Start position	integer	–	Measured	Start position, if subgroup not 0	—
`StartTime` Start time	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
`StartWeight` Start weight	number	kg	Measured	Start weight, if subgroup not 0	—
`StopPosition` Stop position	integer	–	Measured	Stop position, if subgroup not 0	—
`StopTime` Stop time	number	hh-mm-ss	Measured	Stop time, if subgroup not 0	—
`StopWeight` Stop weight	number	kg	Measured	Stop weight, if subgroup not 0	—
`StrawGroup` Straw group	integer	–	—	Straw group	—
`SubGroupNr` Subgroup number	integer	–	—	Number of the subgroup, 0 no subgroup	—
`TakeoffTime` Take off time	number	hh-mm-ss	Measured	Take off time, if log source 1	—
`TrackSection` Track section	integer	–	—	Track section, if subgroup 1 if there is no subgroup 0	—

4.7 GEA separation and smart gate passages

Dataset: `GEA_gates` | **Source table:** `T_RAW_Gea_Separatie` | **Variables:** 5

Table 8: Data dictionary of `GEA_gates`.

Variable	Type	Unit	Origin	Description	Derivation / source
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`event` Event	integer	–	—	0 = No gate action or weighting, 1 = selection gate A, 2 = selection gate B	—
`gate` Gate	integer	–	—	Number of the gate	—
`message` Message	string	tekst	—	ID and info gate action, weighting or selection has or has not taken place	—

4.8 GEA milking transactions

Dataset: `GEA_milkings` | **Source table:** `T_RAW_Gea_milk` | **Variables:** 13

Table 9: Data dictionary of `GEA_milkings`.

Variable	Type	Unit	Origin	Description	Derivation / source
`Cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`IMDat` Gea Date and time	datetime	yy-mm-dd-hh-mm-ss	—	GEA, Date and time milking	—
`Reg1` Registration number	integer	text + number	—	National registration number of the animal.	—
`RespISO` Responder number	integer	–	—	Number of the cows responder	—
`Name` Name	string	—	—	Number in GEA, should be same as cow	—
`IMTim` Gea time	string	—	—	Time?	—
`IMWt` Cow Weight at Milking	number	kg	Measured	Live animal body weight measured by the integrated scale inside the GEA milking stall or exit platform.	Directly recorded by GEA milking scale sensor. *Source:* `T_RAW_Gea_milk`
`IMFlg` Milking Status Flag	string	code	Measured	Status/diagnostic flag from GEA milking robot/parlor indicating normal completion, kick-off, or incomplete milking.	Status code recorded by GEA system. *Source:* `T_RAW_Gea_milk`
`IMDim` Days in Milk	integer	days	Derived	Days in Milk (DIM) recorded in GEA at the time of the milking event.	Milking date minus most recent calving date. *Source:* `T_RAW_Gea_milk`
`IMAvFlo` Avarage milkflow	string	—	—	Average milkflow?	—
`IMMxFlow` Max. milkflow	string	—	—	Max. milkflowAlleen '||||' als waarde???	—
`IMStall` Gea barn	integer	–	—	Place in the Gea milking carousel 1-40	—
`IMStartSec` Gea start sec	string	—	—	Alleen '?????' als waarde???	—

4.9 CowConnect mixer wagon rations

Dataset: `TMR_Trioliet_cowconnect_mix` | **Source table:** `t_raw_cowconnect_mix` | **Variables:** 12

Table 10: Data dictionary of `TMR_Trioliet_cowconnect_mix`.

Variable	Type	Unit	Origin	Description	Derivation / source
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`externalFeedplanID` External feedpland ID	string	—	—	All NULL	—
`feedplanId` Feedplan ID	integer	–	—	Id of the feedplan	—
`feedplanName` Feedplan name	string	text	—	Name of the feedplan	—
`id` ID	string	text	—	ID	—
`isOpen` Is Open	integer	–	—	1 or 0 process running or not.	—
`loadedSum` Loaded som	number	kg	Derived	Total / sum of all loads	—
`mixDate` Mix date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date and time	—
`mixLoaderId` Mix loader ID	integer	–	—	ID of the mix loader	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`mixLoaderName` Mix loader name	string	text	—	Name of the mix loader	—
`rationId` Ration ID	string	—	—	All NULL	—
`type` Type	string	text	—	Type	—

4.10 Nedap Velos behaviour intervals

Dataset: `Velos_behaviour` | **Source table:** `T_RAW_velos_behaviour` | **Variables:** 4

Table 11: Data dictionary of `Velos_behaviour`.

Variable	Type	Unit	Origin	Description	Derivation / source
`date` Date	date	yy-mm-dd	Measured	Date of the data record, UTC timezone	—
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`type` Type	string	text	—	Type of behaviour 1 - 3	0, overig actief. 1, niet actief. 2, herkouwen. 3, vreten.
`duration` Duration	integer	s	Derived	Duration of action/behaviour	Calculated by endTime - startTime

4.11 Nedap Velos daily behaviour budgets

Dataset: `Velos_behaviour_day` | **Source table:** `T_RAW_velos_behaviour` | **Variables:** 7

Table 12: Data dictionary of `Velos_behaviour_day`.

Variable	Type	Unit	Origin	Description	Derivation / source
`date` Date	date	yyyy-mm-dd	Measured	Date of the data record, UTC timezone	—
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`total` Total	number	hh-mm-ss	Derived	Total time recorded of the animal 1 day	Sum duration of all types *Source:* `T_RAW_velos_behaviour`
`overig` Other	number	hh-mm-ss	Derived	Time the cow is actief but not eating or ruminating	Sum duration of type 0 *Source:* `T_RAW_velos_behaviour`
`nietactief` Inactive	number	hh-mm-ss	Derived	Time the cow is inactief	Sum duration of type 1 *Source:* `T_RAW_velos_behaviour`
`herkouwen` Ruminating	number	hh-mm-ss	Derived	Time the cow spent ruminating	Sum duration of type 2 *Source:* `T_RAW_velos_behaviour`
`vreten` Eating	number	hh-mm-ss	Derived	Time the cow spent eating	Sum duration of type 3 *Source:* `T_RAW_velos_behaviour`

4.12 CowConnect feed distribution drops

Dataset: `TMR_Trioliet_cowconnect_drop` | **Source table:** `t_raw_cowconnect_drop` | **Variables:** 11

Table 13: Data dictionary of `TMR_Trioliet_cowconnect_drop`.

Variable	Type	Unit	Origin	Description	Derivation / source
`id` ID	string	text	—	ID	—
`externalMixTaskId` External mix task ID	string	text	—	External mix task ID	—
`quantity` Quantity	number	kg	Measured	Quantity	—
`leftOverQuantity` Left over quantity	number	kg	Measured	Quantity left over	—
`noOfAnimals` Number of animals	integer	–	—	Number of animals	—
`feedSectionName` Feed section name	string	text	—	Name of the feed section	—
`feedsectionType` feed section type	integer	–	—	Feedsection type	—
`externalFeedSectionId` External feed sectiond ID	integer	–	—	External Feed section ID if the feed section has an external ID	—
`externalMixTask` External mix task	string	—	—	—	—
`mix_id` Mix ID	string	text	—	Mix ID and external mix task id are foreign keys of the ID (primairy key)	—
`type` Type	string	text	—	Type	—

4.13 Laboratory dry matter content (drying oven)

Dataset: `DC_DM_content` | **Source table:** `VL_totaal_droogstoof` | **Variables:** 16

Table 14: Data dictionary of `DC_DM_content`.

Variable	Type	Unit	Origin	Description	Derivation / source
`BlkDat` Can date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the can was weighed	—
`BlkGew` Can weight	number	g	Measured	Weight of the can	—
`BlkNum` Can number	integer	–	—	Number of the can	—
`DatMon` Sample date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained	—
`DatSes` Date session	datetime	yy-mm-dd-hh-mm-ss	Measured	Date of weigh session	—
`DrgDat` Dried date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date end of the analysis	—
`DrgGew` Dried weight	number	g	Measured	Dried sample weight, with or without weight of the can?	—
`DrgStfGkg` Dry matter weight kg	number	g	Measured	Weight of the dry matter	—
`DrgTyd` dry time	number	hh-mm-ss	Measured	Time end of the analysis when all moisture has evaporated	—
`ObjCod` Object code	string	text	—	Short description/code of the sample	—
`ObjCodNum` Object code number	integer	–	—	Used when analyzing duplicate or triplicate samples	—
`PrjCod` Project code	string	text	—	Project code	—
`VlgNum` Tracking number	integer	–	—	Tracking number	—
`VrsDat` fresh date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date start of the analysis	—
`VrsGew` fresh weight	number	g	Measured	Weight at the start of the analysis	—
`VrsTyd` fresh time	number	hh-mm-ss	Measured	Time start of the analysis, fresh sample is weighed	—

4.14 GreenFeed enteric emission summaries

Dataset: `Greenfeed_summarized_data` | **Source table:** `T_RAW_Greenfeed_Summarized_Data` | **Variables:** 24

Table 15: Data dictionary of `Greenfeed_summarized_data`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AirflowCf` Airflow correction factor	integer	–	—	Airflow correction factor	—
`AirflowLitersPerSec` Airflow liters per second	integer	$\mathrm{L\,s^{-1}}$	Measured	Airflow liters per second	—
`AnimalName` Animal name	integer	–	Measured	Animal name	—
`CH4GramsPerDay` CH4 grams per day	integer	–	Measured	CH4 grams per day, methane	—
`CO2GramsPerDay` CO2 grams per day	integer	–	Measured	CO2 grams per day, carbon dioxide	—
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`EndTime` End time	number	yy-mm-dd-hh-mm-ss	Measured	End time	—
`GoodDataDuration` Good data duration	datetime	hh-mm-ss	—	Good data duration	—
`greenfeed` Greenfeed	integer	–	Measured	Greenfeed	—
`H2GramsPerDay` H2 grams per day	integer	–	Measured	H2 grams per day, hydrogen	—
`H2SGramsPerDay` H2S grams per day	integer	–	Measured	H2S grams per day, hydrogen sulfide	—
`InterruptingTags` Interrupting tags	integer	–	—	Interrupting tags	—
`IsPreliminary` Is preliminary	integer	–	—	Is preliminary data, validated data comes directly from C-lock.	—
`O2GramsPerDay` O2 grams per day	integer	–	Measured	O2 grams per day, oxygen	—
`RFID` Rfid	integer	–	—	Rfid	—
`RunTime` Run time	number	yy-mm-dd-hh-mm-ss	Measured	Run time	—
`StartTime` Start time	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
`TempPipeDegreesCelsius` Temperature pipe degrees celsius	number	°C	Measured	Temperature pipe degrees celsius	—
`user` User	string	text	—	User	—
`WasInterrupted` Was interrupted	integer	–	—	Was interrupted	—
`WindCf` Wind correction factor	integer	–	—	Wind correction factor	—
`WindDirDeg` Wind direction degrees	integer	–	Measured	Wind direction degrees	—
`WindSpeedMetersPerSec` Wind speed meters per second	integer	$\mathrm{m\,s^{-1}}$	Measured	Wind speed meters per second	—

4.15 GreenFeed bait feed drops

Dataset: `Greenfeed_feed_aggregrated` | **Source table:** `T_RAW_Greenfeed_Feed_data` | **Variables:** 13

Table 16: Data dictionary of `Greenfeed_feed_aggregrated`.

Variable	Type	Unit	Origin	Description	Derivation / source
`cow` Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`CurrentCup` Current cup	integer	–	Measured	Current cup	—
`CurrentPeriod` Current period	integer	–	Measured	Current period	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`DateTime` Date and time	datetime	yyyy-mm-dd hh-mm-ss	Measured	Date and time	—
`DropInterval` Drop interval	integer	s	—	Min. sec between current en new cup	—
`FeedType` Feed type	integer	–	—	Feed type	—
`greenfeed` Greenfeed	integer	–	Measured	Greenfeed	—
`MaxCups` Maximum cups	integer	–	—	Max. cups per period	—
`MaxPeriods` Maximum periods	integer	–	—	Max. periods per day	—
`PeriodInterval` Period interval	integer	s	—	Min. Sec between current en next period	—
`RFIDTag` Rfid tag	integer	–	Measured	Rfid tag	—
`user` User	string	text	—	User	—

4.16 Dairy Campus meteorological station

Dataset: `Weatherstation` | **Source table:** `T_RAW_weerstation` | **Variables:** 18

Table 17: Data dictionary of `Weatherstation`.

Variable	Type	Unit	Origin	Description	Derivation / source
`dewpoint` Dewpoint	number	°C	Measured	Dew point is the temperature to which air must be cooled to become saturated with water vapor	—
`global_radiation` Global radiation	string	$W\,m^{-2}$	Measured	The sum of the diffuse and direct solar radiation is called global solar radiation.	—
`gust` Wind gust	integer	Beaufort, over 10 minuten (meter/sec)	Measured	A sudden, brief increase in speed of the wind	—
`gust_direction` Wind gust direction	string	°	Measured	Direction from which the wind gust blows	—
`humidity_150` Humidity	number	%	Measured	The concentration of water vapor present in the air	—
`latitude` Latitude	string	°	Measured	Coordinate that specifies the north–south position	—
`longitude` Longitude	string	°	Measured	Coordinate that specifies the east–west position	—
`precipitation` Precipitation	string	Millimeters	Measured	Any product of the condensation of atmospheric water vapor that falls from clouds	—
`rain_chance` Rain chance	number	%	Measured	Chance of precipitation	—
`station_name` Station name	string	text	—	Station name	—
`symbol_number` Symbol number	string	—	—	Symbol number	—
`temperature_10` Temperature 10	number	°C	Measured	Temperature 10 cm above ground level.	—
`temperature_150` Temperature 150	number	°C	Measured	Temperature 150 cm above ground level.	—
`timestamp` Timestamp	datetime	yy-mm-dd-hh-mm-ss	Measured	Timestamp	—
`wet_bulb_temperature` Wet bulb temperature	number	°C	Measured	Thermometer with the bulb in somthing wet	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`wind_direction` Wind direction	string	°	Measured	Direction from which the wind blows	—
`windchill` Windchill	number	°C	Measured	The cooling effect of wind blowing on a surface	—
`windspeed` Windspeed	integer	Beaufort, over 10 minuten (meter/sec)	Measured	Rate at which wind/air is moving	—

4.17 Data completeness audit matrix

Dataset: `Missing_data` | **Source table:** `tvl_totaal_missende_data_proef` | **Variables:** 17

Table 18: Data dictionary of `Missing_data`.

Variable	Type	Unit	Origin	Description	Derivation / source
`bcs` Body condition score	boolean	yes/no	Measured	Automatic body condition score camera	number
`behcod` Treatment code	string	yes/no	Measured	Treatment code	—
`bijzonderheden` Particularities	boolean	yes/no	Measured	Particularities	—
`date` Date	date	yy-mm-dd	Measured	Date of the data record	—
`gewa` Weight evening	boolean	yes/no	Measured	Weight evening	kg
`gewgeaa` Weight gea evening	boolean	yes/no	Measured	Weight gea evening	kg
`gewgeao` Weight gea morning	boolean	yes/no	Measured	Weight gea morning	kg
`gewo` Weight morning	boolean	yes/no	Measured	Weight morning	kg
`hok` Barn	boolean	yes/no	Measured	Barn	number
`kgmlka` Kg milk evening	boolean	yes/no	Measured	Kg milk evening	kg
`kgmlko` Kg milk morning	boolean	yes/no	Measured	Kg milk morning	kg
`oornum` Ear number	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
`opnkv` Intake feed	boolean	yes/no	Measured	Intake feed	kg
`opnrv` Intake roughage	string	yes/no	Measured	Intake roughage	kg
`prdsta` Production stage	boolean	yes/no	Measured	Identifying the stage of the cow, producing milk or not	short text
`prfcod` Trial code	string	yes/no	Measured	Trial code	—
`velos` Velos	boolean	yes/no	Measured	Velos cow behavior	—

4.18 Barn environment and animal daily profile

Dataset: `Environment_barn_Animal_day` | **Source table:** `TVL_Dier_Datum_Stal` | **Variables:** 52

Table 19: Data dictionary of `Environment_barn_Animal_day`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AgeDAys` Age	integer	–	Measured	Age of the animal in days	—
`AnimalNumber` Animal Number	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`AntGewA` Weight afternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
`AntGewO` Weight morning	number	kg	Measured	Weight of the animal after milking in the morning	—
`BirthDate` Date of birth	date	yyyy-mm-dd	Measured	The cows date of birth	—
`DateMelkCon4Wk` Date of 4-Weekly Milk Recording	date	yyyy-mm-dd	Measured	Date of the most recent official 4-weekly milk control / recording event (MPR).	Date of official test day sampling. *Source:* `T_Animal_Milk Samples`
`DatGldTot` Valid Until Date	datetime	yyyy-mm-dd	Derived	Date until which the current housing, lactation, or trial assignment status remains valid.	End validity date of status record. *Source:* `TVL_Dier_ Datum_Stal`
`DaysInLactation` Days in lactation	integer	–	Derived	Amount of days producing milk	Date - calving date
`DryDate` Start date dry period	date	yy-mm-dd	Measured	Starting date of the dry period	—
`EarNumber` Ear number	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
`Eiw` Protein concentration standard	number	%	Derived	Protein concentration of the monthly milk test	—
`EndDate` Ending date experiment	date	yy-mm-dd	Measured	Expected end date of the experiment	—
`Ket` Ketose concentration of the monthly milk test	string	Calculated	Derived	number	Laboratory diagnostic score from milk recording analysis. *Source:* `T_Animal_Milk Samples`
`ExpecCalvingDate` Expected calving date	date	yy-mm-dd	Derived	Expected calving date based on insemination date	Calculated by ExpectedCalvingDate() function from last successful insemination. *Source:* `t_animal_ matings / Expected CalvingDate`
`Fat%` Calculated day fat concentration	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	—
`GewTot` Weight totaal	number	kg	Measured	Average weight of the animal for the day.	Recorded by automated walkthrough weighing scale. *Source:* `TVL_Dier_ Gewicht_Etmaal`
`IDNumber` ID nummer	integer	National registration number of the animal.	Measured	Registration number same as LevNum	Official national cattle ear tag number. *Source:* `t_animals`
`InsiminationDate` Insemination day	date	yy-mm-dd	Measured	Date of last insemination	—
`Lac` Lactose concentration standard	number	%	Derived	Lactose of the monthly milk test	—
`Lactose%` Calculated day lactose concentration	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	—
`LastCalvingDate` Last calving date	date	yy-mm-dd	Measured	Date of last calving	—
`Milk_Yield` Milk Yield	number	kg	Derived	Milk yield	—
`MilkAfternoon` Milk Yield afternoon	number	kg	Derived	Milk Yield afternoon	—
`MilkDay` Milk Yield day	number	kg	Derived	Milk Yield day	—
`MilkMorning` Milk Yield morning	number	kg	Derived	Milk Yield morning	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`MilkYield` Milk Yield	number	kg	Derived	Milk yield	—
`MlkCon` Milk Con?	boolean	boolean	Derived	Indicator flag showing whether official milk recording (MPR) data was collected on this date.	Boolean flag indicating presence of milk laboratory sample record. *Source:* `T_Animal_Milk Samples`
`NumberAnimal` Number Animal	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
`NumberOLactations` Parity	integer	–	Measured	Parity (amount of calvings) of the animal	—
`NumbOfDates` Number of dates?	date	count	Derived	Count of observation days recorded for the animal within the experimental reporting period.	Count of distinct active recording dates. *Source:* `TVL_Dier_ Datum_Proef`
`Protein%` Calculated day protein concentration	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	—
`SampleDateEvening` Sample date evening	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the evening	—
`SampleDateMorning` Sample date morning	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the morning	—
`SCC` Calculated day somatic cell count	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	—
`StartDate` Start date	datetime	hh-mm-ss	Measured	Start date	—
`Stl` Barn Section	string	identifier	Measured	Physical barn building or housing section identifier at Dairy Campus.	Stable / Barn number recorded in management system. *Source:* `TVL_Dier_ Datum_Stal`
`SumVoeGft` Total Feed Granted	number	kg	Derived	Total amount of concentrate feed programmed/granted to the animal over the period.	Sum of programmed concentrate feeding amounts. *Source:* `TVL_Dier_ Voeropname_KV_Etmaal`
`SumVoeOpnm` Total Feed Intake	number	kg	Derived	Total actual concentrate feed consumed by the animal over the period.	Sum of consumed concentrate intake amounts. *Source:* `TVL_Dier_ Voeropname_KV_Etmaal`
`SumVoeRst` Total Feed Residual	number	kg	Derived	Total amount of unconsumed concentrate feed remaining in the troughs.	Total Feed Granted minus Total Feed Intake. *Source:* `TVL_Dier_ Voeropname_KV_Etmaal`
`Unit` Unit / barn number	integer	–	—	Location of where the cows are housed.	—
`Ure` Calculated day urea concentration	string	mg dL^{-1}	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	—
`Urea (mg/100g)` Weekly milk urea concentration	string	mg dL^{-1}	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`Vet` Weekly milk fat concentration	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`Week` Week	integer	–	Measured	Calender week number of data record	—
`WeekDay` Day of the week	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—
`WeightAfternoon` Weight afternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
`WeightMorning` Weight morning	number	kg	Measured	Weight of the animal after milking in the morning	—
`Year` number	string	—	—	Measured	—
`VrwDgn` Expected Days	integer	days	Derived	Expected remaining gestation days or planned days dry prior to the anticipated calving date.	Expected calving date minus current date. *Source:* `TVL_Dier_Datum_Stal`
`AntMlkO` Number of milkings morning	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`AntMlkA` Number of milkings afternoon	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`Date` Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—

4.19 Experimental cohort animal daily records

Dataset: `Environment_barn_Animal_experim` | **Variables:** 54

Table 20: Data dictionary of `Environment_barn_Animal_experim`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AntMlkA` Number of milkings afternoon	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`AntMlkO` Number of milkings morning	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`Date` Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
`VrwDgn` Expected Days	integer	days	Derived	Expected remaining gestation days or planned days dry prior to the anticipated calving date.	Expected calving date minus current date. *Source:* `TVL_Dier_Datum_Stal`
`AgeDays` Age in Days	integer	days	Derived	Age of the animal in days calculated as recording date minus birth date.	Recording date - Birth date. *Source:* `TVL_Dier_Datum_Stal`
`AnimalNumber` Animal Number	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`BirthDate` Date of birth	date	yyyy-mm-dd	Measured	The cows date of birth	—
`DaysInLactation` Days in lactation	integer	–	Derived	Amount of days producing milk	Date - calving date
`DryDate` Start date dry period	date	yy-mm-dd	Measured	Starting date of the dry period	—
`EarNumber` Ear number	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
`EndDate` Ending date experiment	date	yy-mm-dd	Measured	Expected end date of the experiment	—
`Fat%` Calculated day fat concentration	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	—
`GewTot` Total Weight	number	kg	Measured	Total live animal body weight (average or latest measurement for the date).	Recorded by automated walkthrough weighing scale. *Source:* `TVL_Dier_Gewicht_Etmaal`
`InsiminationDate` Insemination day	date	yy-mm-dd	Measured	Date of last insemination	—
`Ket` Ketosis Indicator	string	code	Derived	Ketosis classification indicator (0: healthy/negative, 1: suspected subclinical ketosis, 2: clinical ketosis).	Laboratory diagnostic score from milk recording analysis. *Source:* `T_Animal_Milk Samples`
`Lactose%` Calculated day lactose concentration	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	—
`LastCalvingDate` Last calving date	date	yy-mm-dd	Measured	Date of last calving	—
`NumberAnimal` Number Animal	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
`NumberOLactations` Parity	integer	–	Measured	Parity (amount of calvings) of the animal	—
`NumbOfDates` Number of Dates	date	count	Derived	Count of observation days recorded for the animal within the experimental reporting period.	Count of distinct active recording dates. *Source:* `TVL_Dier_Datum_Proef`
`Projectcode` Project Code	string	identifier	Measured	Unique research experiment / trial code (e.g. 4400003634). Same as prfcod.	Official trial identifier registered in t_proef. *Source:* `t_proef`
`ProjectName` Project Name	string	text	Measured	Official descriptive name of the research project or trial. Same as prfnam.	Official trial title registered in t_proef. *Source:* `t_proef`
`Protein%` Calculated day protein concentration	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	—
`SampleDateEvening` Sample date evening	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the evening	—
`SampleDateMorning` Sample date morning	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the morning	—
`SCC` Calculated day somatic cell count	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	—
`StartDate` Start date	datetime	hh-mm-ss	Measured	Start date	—
`SumVoeGft` Total Feed Granted	number	kg	Derived	Total amount of concentrate feed programmed/granted to the animal over the period.	Sum of programmed concentrate feeding amounts. *Source:* `TVL_Dier_Voeropname_KV_Etmaal`

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`SumVoeRst` Total Feed Residual	number	kg	Derived	Total amount of unconsumed concentrate feed remaining in the troughs.	Total Feed Granted minus Total Feed Intake. *Source:* `TVL_Dier_Voeropname_KV_Etmaal`
`Unit` Unit / barn number	integer	–	—	Location of where the cows are housed.	—
`Urea (mg/100g)` Weekly milk urea concentration	string	$\mathrm{mg\,dL^{-1}}$	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`Week` Week	integer	–	Measured	Calender week number of data record	—
`WeekDay` Day of the week	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—
`WeightAfternoon` Weight afternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
`WeightMorning` Weight morning	number	kg	Measured	Weight of the animal after milking in the morning	—
`Year` number	string	—	—	Measured	—
`AntGewA` Weight afternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
`AntGewO` Weight morning	number	kg	Measured	Weight of the animal after milking in the morning	—
`IDNumber` Animal Identification Number	string	identifier	Measured	Official national unique life identification number of the animal (LevNum / EU identification tag).	Official national cattle ear tag number. *Source:* `t_animals`
`DateMelkCon4Wk` Date of 4-Weekly Milk Recording	date	yyyy-mm-dd	Measured	Date of the most recent official 4-weekly milk control / recording event (MPR).	Date of official test day sampling. *Source:* `T_Animal_Milk Samples`
`DatGldTot` Valid Until Date	datetime	yyyy-mm-dd	Derived	Date until which the current housing, lactation, or trial assignment status remains valid.	End validity date of status record. *Source:* `TVL_Dier_Datum_Stal`
`Eiw` Protein concentration standard	number	%	Derived	Protein concentration of the monthly milk test	—
`ExpecCalvingDate` Expected Calving Date	date	yyyy-mm-dd	Derived	Estimated date of the next calving, based on confirmed insemination date + gestation length (~270-280 days).	Calculated by ExpectedCalvingDate() function from last successful insemination. *Source:* `t_animal_matings / Expected CalvingDate`
`Lac` Lactose concentration standard	number	%	Derived	Lactose of the monthly milk test	—
`Milk_Yield` Milk Yield	number	kg	Derived	Milk yield	—
`MilkAfternoon` Milk Yield afternoon	number	kg	Derived	Milk Yield afternoon	—
`MilkDay` Milk Yield day	number	kg	Derived	Milk Yield day	—
`MilkMorming` Morning Milk Yield	number	kg	Measured	Total quantity of milk produced during morning milking sessions.	Sum of morning milking yields. *Source:* `TVL_Dier_Melkgift_Etmaal_Melkcontrole`
`MilkYield` Milk Yield	number	kg	Derived	Milk yield	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
`MlkCon` Milk Recording Present	boolean	boolean	Derived	Indicator flag showing whether official milk recording (MPR) data was collected on this date.	Boolean flag indicating presence of milk laboratory sample record. *Source:* `T_Animal_Milk Samples`
`Stl` Barn Section	string	identifier	Measured	Physical barn building or housing section identifier at Dairy Campus.	Stable / Barn number recorded in management system. *Source:* `TVL_Dier_ Datum_Stal`
`Ure` Calculated day urea concentration	string	$\mathrm{mg\,dL^{-1}}$	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	—
`Vet` Weekly milk fat concentration	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
`SumVoeOpnm` Total Feed Intake	number	kg	Derived	Total actual concentrate feed consumed by the animal over the period.	Sum of consumed concentrate intake amounts. *Source:* `TVL_Dier_ Voeropname_KV_Etmaal`

5 Data availability and citation

The machine-readable dictionary (CSV, XLSX), per-dataset CSV files and the SQL calculation layer (views, stored procedures and functions) are deposited with this descriptor. Please cite:

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